

INNOVATION

University Hackathon

Life Is On

Schneider
Electric

Team Name : DigiSeva

Team Member-1
DEVANSH GOENKA

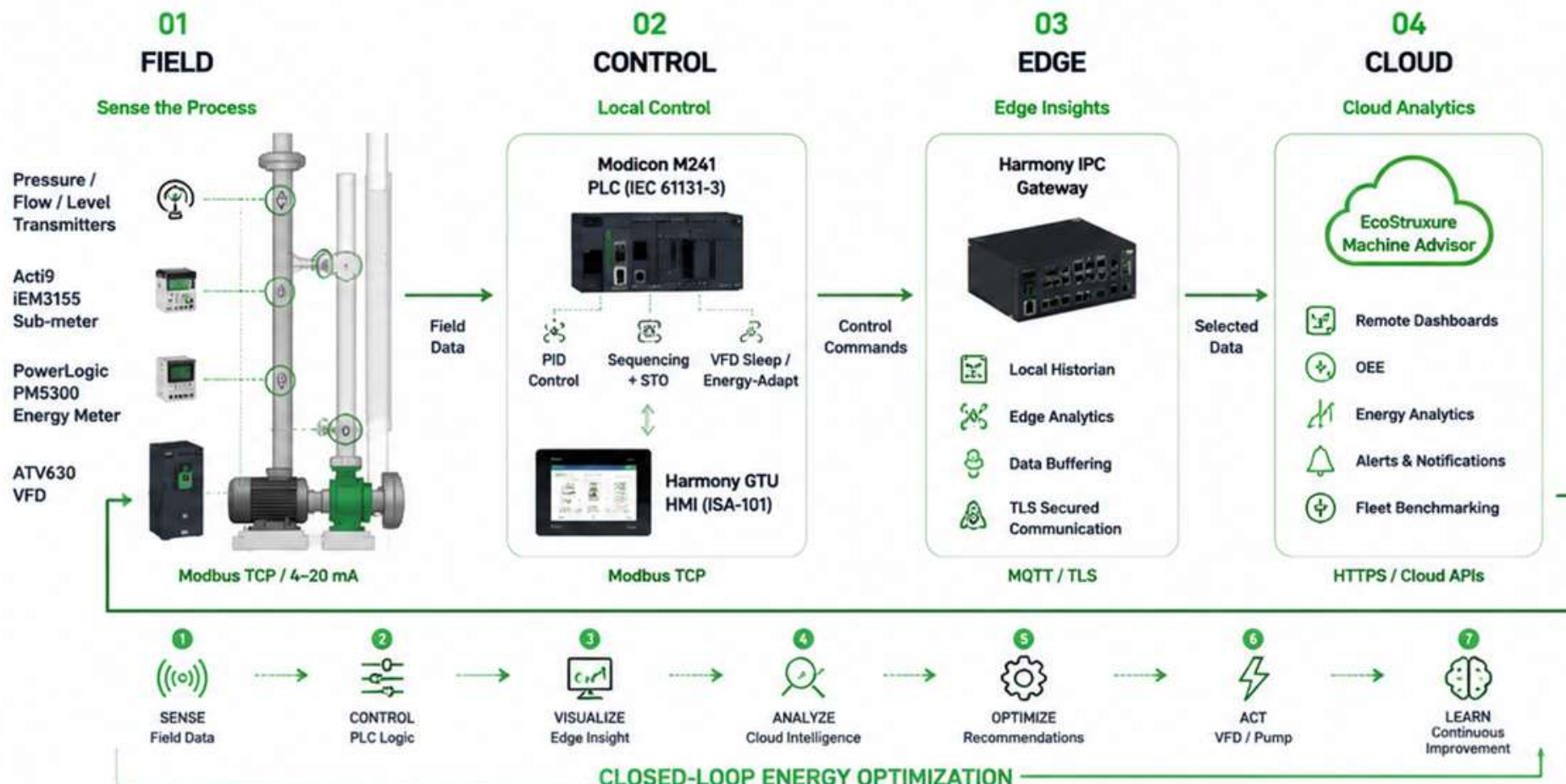
Team Member-2
PANSHUL ARORA

Team Member-3
TANMAY SINGH

Architecture / Approach

Field → Control → Edge → Cloud

Closed-loop energy optimization for a 30 kW variable-demand water pumping station



- ### 01 APPROACH
- 01** Replace valve throttling with VFD speed control.
 - 02** Use pump affinity / cube-law behavior: $P \propto N^3$.
 - 03** Calibrate savings against the actual system curve: static head + friction head.
 - 04** Continuously: SENSE → CONTROL → VISUALIZE → ANALYZE → OPTIMIZE → ACT → LEARN

VALUE
DELIVERED



~46%
ENERGY REDUCTION



0.58 → 0.31
kWh/m³ SEC
SPECIFIC ENERGY CONSUMPTION



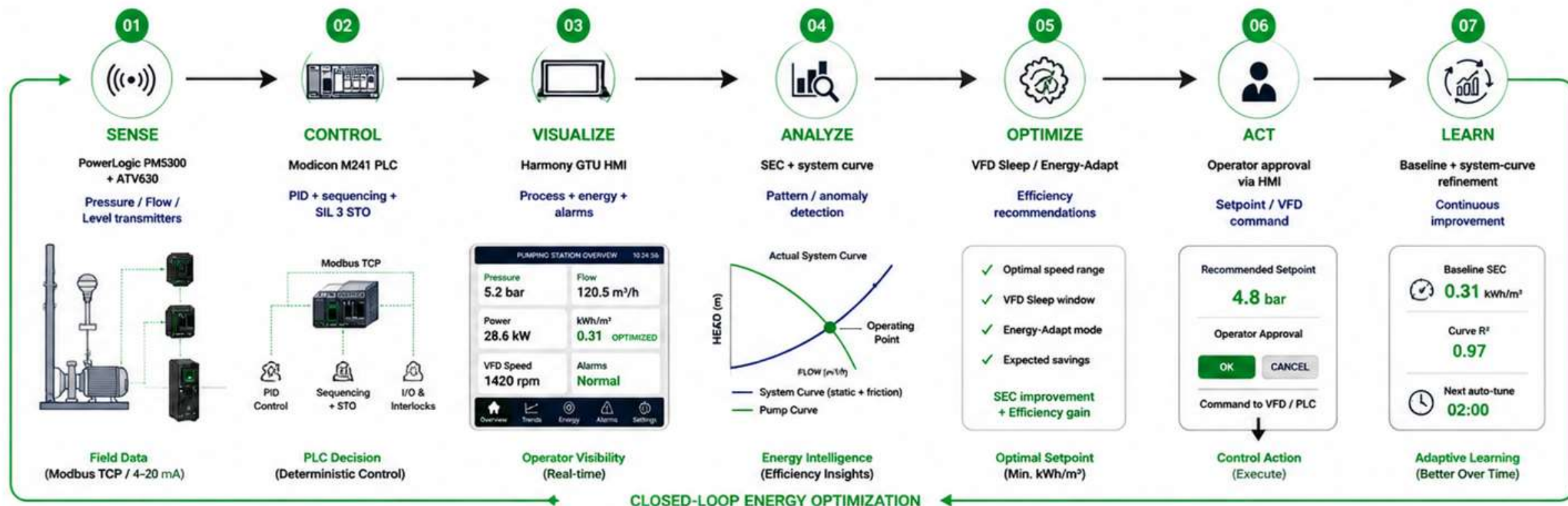
₹4-6
LAKH / YEAR
ENERGY SAVED



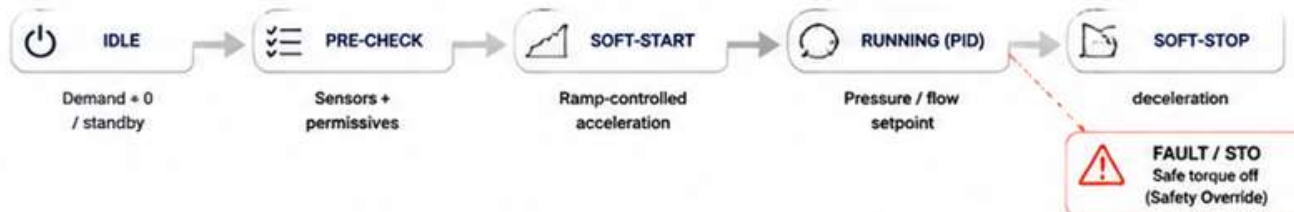
< 2 YEARS
PAYBACK

Overall Sequence flow

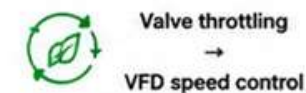
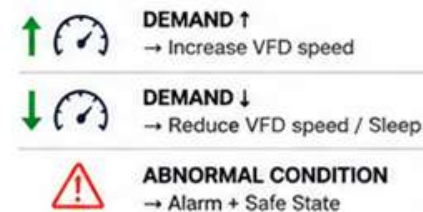
> LIVE HMI PROTOTYPE
dg10911.github.io/ecodrive-hmi



PLC CONTROL SEQUENCE



DECISION LOGIC



$$P \propto N^3$$

WHY IT MATTERS



~46%
ENERGY REDUCTION



0.58 → 0.31
kWh/m³ SEC
SPECIFIC ENERGY CONSUMPTION



₹4-6
LAKH / YEAR
ENERGY SAVINGS



< 2 YEARS
PAYBACK

Tools used



SCHNEIDER HARDWARE

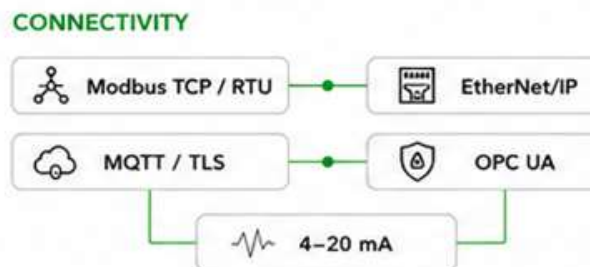
- MODICON M241**
PLC • IEC 61131-3
 - HARMONY GTU**
HMI • ISA-101
 - ALTIVAR PROCESS ATV630**
VFD • Sleep Mode • Energy-Adapt • STO
 - POWERLOGIC PM5300**
Energy Meter
 - ACTI9 iEM3155**
Sub-meter
 - PRESSURE / FLOW / LEVEL**
Transmitters
 - CONNEXIUM**
Managed Ethernet Switch
- Sensors → M241 PLC → ATV630 VFD → Motor / Pump



SOFTWARE & COMMUNICATION



- ### CONTROL & ENGINEERING
- EcoStruxure Machine Expert**
IEC 61131-3: ST / FBD / SFC / LD
 - Vijeo Designer / Operator**
Terminal Expert (HMI)



- ### EDGE & CLOUD
- EcoStruxure Machine Advisor**
IoT / Cloud
 - Node-RED**
 - Grafana**
 - Raspberry Pi / ESP32**
Prototype / Edge



STANDARDS & ENGINEERING



ISA-101
High-Performance HMI



ISA-18.2 / IEC 62682
Alarm Management



IEC 61131-3
PLC Programming



IEC 61508
Functional Safety
Safe Torque Off



IEEE 519
Power Quality / THD



ISO 50001
Energy Management



IEC 62443
OT Cybersecurity



FIELD



CONTROL



HMI



EDGE



CLOUD

Sensors + PLC + VFD + HMI + Edge + Cloud



INDUSTRIAL

Built on Schneider Electric hardware



OPEN

Open standards & open edge stack



STANDARDS-BASED

Safety • Energy • Cybersecurity ready